SampleNet: Contrastive Distillation for Neural Paper Summarization

Jane Doe, John Smith - Institute of Sample Research

Abstract

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SampleNet processes a paper in three stages. First, the structure-aware encoder reads each section separately and tags tokens with their section type, such as abstract, introduction, method, or experiment, producing contextual embeddings that know where each sentence comes from. Second, the gated sentence selector computes a saliency gate g for every sentence, and the contrastive objective pulls the summary representation close to sentences describing the core contribution while pushing away background sentences. The training loss is L = L_ce + lambda * L_cl, where L_ce is the standard cross-entropy

of summary generation and L_cl is the contrastive term; we set lambda to 0.5 after a small grid search. Third, the distillation stage uses a large teacher reader to produce rationales, short explanations of why each selected sentence matters, and trains the student to imitate both the summaries and the rationales. The student is 12 times smaller than the teacher. We hypothesize this works because rationales force the student to attend to causal parts of the paper rather than to frequent surface phrases, and the contrastive term prevents collapse onto boilerplate.

4 Experiments

We evaluate on three benchmarks: arXiv-Abst, PubMed-Intro, and SciSumm-100, covering papers from machine learning and biomedicine. Baselines are Lead-3, TextRank, BertSumExt, and a large language model prompted with a single-shot instruction. Main results: SampleNet achieves ROUGE-L 41.7 on arXiv-Abst, versus 37.5 for the best baseline; on PubMed-Intro the gain is 3.1 points; on SciSumm-100 the gain is 2.8 points. In human evaluation with three expert annotators, SampleNet summaries are preferred in 68 percent of pairwise comparisons, mainly for faithfulness and coverage of the contribution. Training details: we train for 60,000 steps with AdamW, learning rate 3e-5, batch size 32, on four A100 GPUs, taking about 14 hours; inference takes 1.2 seconds per paper on a single GPU. Ablation study: removing the contrastive term drops ROUGE-L by 2.4 points; removing section-aware encoding drops it by 1.6 points; removing rationale distillation drops human preference by 11 points, the largest single effect. We observe that the gains concentrate on papers with long related-work sections, which suggests that structure awareness, not raw capacity, drives most of the improvement. A limitation is that SampleNet still copies numbers inaccurately in about 6 percent of summaries.

5 Conclusion

We presented SampleNet, a framework that combines structure-aware encoding, contrastive selection, and rationale distillation for paper summarization. It outperforms strong baselines on three benchmarks and produces summaries that expert readers prefer. Future work includes extending the approach to multi-document summaries and to papers outside computer science.

SampleNet: Contrastive Distillation for Neural Paper Summarization

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Abstract

Automatic summarization of scientific papers remains difficult because models tend to copy surface text instead of capturing the core contribution. We propose SampleNet, a summarization framework that combines contrastive learning with knowledge distillation from a large teacher reader. SampleNet first encodes each section with a structure-aware encoder, then selects salient sentences through a learned gate, and finally distills the teacher's rationale into a compact summary. On three benchmarks, SampleNet improves ROUGE-L by 4.2 points over the strongest baseline, and human evaluators prefer its summaries in 68 percent of cases.

1 Introduction

Scientific papers are growing in number and length, making it hard for researchers to keep up with their fields. Summarization systems can help, but existing neural summarizers often produce generic text that misses the key contribution of a paper. Prior work falls into two families. Extractive methods select sentences by salience scores; they are faithful but disjoint. Abstractive methods generate fluent text but hallucinate facts and ignore section structure. Neither family explains why a result matters, which is exactly what readers care about. In this work we ask: can a model learn to summarize like an expert reader, emphasizing the problem, the method, and the evidence in proportion? Our answer is SampleNet, which has three components: a structure-aware section encoder, a gated sentence selector, and a distillation stage that transfers the rationale of a large teacher reader into a small student model. Our contributions are: (1) a structure-aware encoding scheme for papers; (2) a contrastive objective that separates core contributions from background; and (3) a distillation recipe that yields a 12x smaller model with better summaries.

2 Related Work

Extractive summarization dates back to salience scoring with graph-based ranking, and recent neural variants score sentences with contextual embeddings. Abstractive summarization advanced with sequence-to-sequence attention models and large pretrained language models, which improved fluency but increased hallucination rates. Distillation has been used to compress language models for efficiency, but using a teacher reader to transfer summarization rationale is, to our knowledge, new.

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